



# SAFETY DATA SHEET

This safety data sheet was created pursuant to the requirements of:  
GB/T 16483-2008, GB/T 17519-2013

**Product Name** Liquichek Cardiac Advance Control

**Revision date** 18-Mar-2025

**Revision Number** 2

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### Product identifier

**Product Name** Liquichek Cardiac Advance Control

**Catalogue Number(s)** 12016447, 12016448, 12016449, 12016450, 12016451, 12016452, 12016453, 12016454, 12016472, 12016473, 12016474, 12016475

### Other means of identification

**Pure substance/mixture** Mixture

### Details of the supplier of the safety data sheet

#### Corporate Headquarters

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories Inc.  
9500 Jeronimo Road  
Irvine, California 92618  
USA

#### Legal Entity / Contact Address

Bio-Rad Laboratories Ltd.  
1st and 2nd Floor, Lumpini 1 Building  
239/2, Rajdamri Road, Lumpini,  
Pathumwan, Bangkok 10330  
Thailand

**Technical Service** +66 2 652 8313  
ctsthailand@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC Hong Kong: 800-968-793

### Recommended use of the chemical and restrictions on use

**Recommended use** In vitro diagnostic

## SECTION 2: Hazards identification

### **Emergency Overview**

May cause allergic skin reaction

**Appearance** Clear

**Physical state** Liquid

**Odour** Mild

### Classification of the substance or mixture

|                                                              |            |
|--------------------------------------------------------------|------------|
| <b>Skin sensitisation</b>                                    | Category 1 |
| <b>Hazardous to the Aquatic Environment - Acute Hazard</b>   | Category 3 |
| <b>Hazardous to the Aquatic Environment - Chronic Hazard</b> | Category 3 |

### Label elements

**Signal word**

Warning

**Hazard statements**

May cause an allergic skin reaction  
Harmful to aquatic life with long lasting effects

**Precautionary statements****Prevention**

Wear protective gloves/protective clothing/eye protection/face protection  
Avoid release to the environment  
IF ON SKIN: Wash with plenty of water and soap  
If skin irritation or rash occurs: Get medical advice/attention

**Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Physical and chemical hazards**

Not applicable.

**Health hazards**

Immediate Health Effects: Allergic reactions (sensitiser). Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain, or flushing.  
Chronic effects: Not applicable.

**Environmental hazards**

This material is a water pollutant. Keep out of drains, sewers, ditches and waterways. Minimise use of water to prevent environmental contamination

**Other hazards which do not result in classification**

Contains animal source material (Cattle) Contains animal source material (Cattle) (fish)

Contains human source material and / or potentially infectious components.

**SECTION 3: Composition/information on ingredients****Substance**

Not applicable.

**Mixture**

| Chemical name                                                                    | Weight-%     | CAS No.    |
|----------------------------------------------------------------------------------|--------------|------------|
| Potassium chloride                                                               | 1 - 2.5      | 7447-40-7  |
| Sodium thiocyanate                                                               | 0.1 - 0.249  | 540-72-7   |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone | 0.001 - 0.01 | 55965-84-9 |

**SECTION 4: First aid measures**

**Description of necessary first aid measures**

|                                                                           |                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b>                                                     | Show this safety data sheet to the doctor in attendance.                                                                                                                                                                  |
| <b>Inhalation</b>                                                         | Remove to fresh air.                                                                                                                                                                                                      |
| <b>Eye contact</b>                                                        | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a doctor. Call a doctor. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. |
| <b>Skin contact</b>                                                       | Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation or allergic reactions see a doctor.                                                                                         |
| <b>Ingestion</b>                                                          | Call a doctor.                                                                                                                                                                                                            |
| <b><u>Most important symptoms and effects, both acute and delayed</u></b> | Itching. Rashes. Hives.                                                                                                                                                                                                   |
| <b><u>For emergency responders</u></b>                                    | No information available.                                                                                                                                                                                                 |
| <b><u>Note to doctors</u></b>                                             | May cause sensitisation in susceptible persons. Treat symptomatically.                                                                                                                                                    |

**SECTION 5: Firefighting measures****Extinguishing media**

|                                                            |                                                                                                                                    |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>                        | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                            |
| <b>Unsuitable extinguishing media</b>                      | No information available.                                                                                                          |
| <b><u>Specific hazards arising from the chemical</u></b>   | Product is or contains a sensitizer. May cause sensitisation by skin contact.                                                      |
| <b><u>Special protective actions for fire-fighters</u></b> | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. Use personal protection equipment. |

**SECTION 6: Accidental release measures****Personal precautions, protective equipment and emergency procedures**

|                                                                    |                                                                                                                                                                                                          |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal precautions</b>                                        | Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. |
| <b>For emergency responders</b>                                    | Use personal protection recommended in Section 8.                                                                                                                                                        |
| <b><u>Environmental precautions</u></b>                            | See Section 12 for additional Ecological Information.                                                                                                                                                    |
| <b><u>Methods and material for containment and cleaning up</u></b> | Prevent further leakage or spillage if safe to do so. Clean contaminated surface thoroughly.                                                                                                             |
| <b><u>Precautions to prevent secondary hazards</u></b>             | Clean contaminated objects and areas thoroughly observing environmental regulations.                                                                                                                     |

**SECTION 7: Handling and storage**

|                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>Precautions for safe handling</u></b>                                | Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash it before reuse. See Section 8 for information on appropriate personal protective equipment. |
| <b><u>Conditions for safe storage, including any incompatibilities</u></b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Store locked up. Keep out of the reach of children.                                                                                                                                                                                                                                                                                      |
| <b><u>Incompatible materials</u></b>                                       | None known based on information supplied.                                                                                                                                                                                                                                                                                                                                                                         |

**SECTION 8: Exposure controls/personal protection****Occupational exposure limits**

This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Note** See section 16 for terms and abbreviations

**Biological occupational exposure limits**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Monitoring and observation processes**

No applicable information was found.

|                                    |                                                     |
|------------------------------------|-----------------------------------------------------|
| <b><u>Engineering controls</u></b> | Showers<br>Eyewash stations<br>Ventilation systems. |
|------------------------------------|-----------------------------------------------------|

**Individual protection measures, such as personal protective equipment**

|                                              |                                                                                                                                                                          |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><u>Eye/face protection</u></b>            | Wear safety glasses with side shields (or goggles).                                                                                                                      |
| <b><u>Skin and body protection</u></b>       | Wear suitable protective clothing.                                                                                                                                       |
| <b><u>Hand protection</u></b>                | Wear suitable gloves.                                                                                                                                                    |
| <b><u>Respiratory protection</u></b>         | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
| <b><u>General hygiene considerations</u></b> | Handle in accordance with good industrial hygiene and safety practice.                                                                                                   |

**SECTION 9: Physical and chemical properties****Information on basic physical and chemical properties**

|                               |                          |
|-------------------------------|--------------------------|
| <b><u>Appearance</u></b>      | Clear                    |
| <b><u>Colour</u></b>          | light yellow             |
| <b><u>Physical state</u></b>  | Liquid                   |
| <b><u>Odour</u></b>           | Mild                     |
| <b><u>Odour threshold</u></b> | No information available |

| <u>Property</u>                         | <u>Values</u>     | <u>Remarks • Method</u> |
|-----------------------------------------|-------------------|-------------------------|
| pH                                      | 7.0-7.3           |                         |
| Melting point / freezing point          | No data available | None known              |
| Initial boiling point and boiling range | No data available | None known              |
| Flash point                             | No data available | None known              |
| Evaporation rate                        | No data available | None known              |
| Flammability                            | No data available | None known              |
| Flammability Limit in Air               |                   | None known              |
| Upper flammability or explosive limits  | No data available |                         |
| Lower flammability or explosive limits  | No data available |                         |
| Vapour pressure                         | No data available | None known              |
| Relative vapour density                 | No data available | None known              |
| Relative density                        | No data available | None known              |
| Water solubility                        | Miscible in water |                         |
| Solubility(ies)                         | No data available | None known              |
| Partition coefficient                   | No data available | None known              |
| Autoignition temperature                | 100 °C            |                         |
| Decomposition temperature               | No data available | None known              |
| Kinematic viscosity                     | No data available | None known              |
| Dynamic viscosity                       | No data available | None known              |
| <u>Additional information</u>           |                   |                         |
| Explosive properties                    | Not applicable    |                         |
| Oxidising properties                    | Not applicable    |                         |

## SECTION 10: Stability and reactivity

|                                           |                                           |
|-------------------------------------------|-------------------------------------------|
| <u>Stability</u>                          | Stable under normal conditions.           |
| <u>Possibility of hazardous reactions</u> | None under normal processing.             |
| Sensitivity to mechanical impact          | None.                                     |
| Sensitivity to static discharge           | None.                                     |
| <u>Conditions to avoid</u>                | None known based on information supplied. |
| <u>Incompatible materials</u>             | None known based on information supplied. |
| <u>Hazardous decomposition products</u>   | None known based on information supplied. |

## SECTION 11: Toxicological information

### Acute toxicity

### Numerical measures of toxicity

No information available

The following values are calculated based on chapter 3.1 of the GHS document

|                 |                  |
|-----------------|------------------|
| ATEmix (oral)   | 111,995.80 mg/kg |
| ATEmix (dermal) | 65,573.80 mg/kg  |

### Component Information

| Chemical name      | Oral LD50            | Dermal LD50 | Inhalation LC50 |
|--------------------|----------------------|-------------|-----------------|
| Potassium chloride | = 2600 mg/kg ( Rat ) | -           | -               |

|                                                                                  |                     |                          |   |
|----------------------------------------------------------------------------------|---------------------|--------------------------|---|
| Sodium thiocyanate                                                               | = 764 mg/kg ( Rat ) | -                        | - |
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone | = 53 mg/kg ( Rat )  | = 87.12 mg/kg ( Rabbit ) | - |

**Skin corrosion/irritation** Based on available data, the classification criteria are not met.

**Serious eye damage/eye irritation** Based on available data, the classification criteria are not met.

**Respiratory or skin sensitisation** May cause sensitisation by skin contact.

**Germ cell mutagenicity** Based on available data, the classification criteria are not met.

#### Carcinogenicity

The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Chemical name      | China | IARC     |
|--------------------|-------|----------|
| Sodium thiocyanate | -     | Group 2A |

#### Legend

**IARC (International Agency for Research on Cancer)**

Group 2A - Probably Carcinogenic to Humans

**Reproductive toxicity** Based on available data, the classification criteria are not met.

**Specific target organ toxicity — single exposure** Based on available data, the classification criteria are not met.

**Specific target organ toxicity — repeated exposure** Based on available data, the classification criteria are not met.

**Target organ effects** Respiratory system. Eyes.

**Aspiration hazard** Based on available data, the classification criteria are not met.

## SECTION 12: Ecological information

**Ecotoxicity** Harmful to aquatic life with long lasting effects.

| Chemical name      | Algae/aquatic plants                                   | Fish                                                                                                          | Crustacea                                                                                 |
|--------------------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Potassium chloride | EC50: =2500mg/L (72h, <i>Desmodesmus subspicatus</i> ) | LC50: =1060mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: 750 - 1020mg/L (96h, <i>Pimephales promelas</i> ) | EC50: =825mg/L (48h, <i>Daphnia magna</i> )<br>EC50: =83mg/L (48h, <i>Daphnia magna</i> ) |

**Persistence and degradability** No information available.

**Bioaccumulative potential** There is no data for this product.

#### Component Information

| Chemical name                                                                    | Partition coefficient |
|----------------------------------------------------------------------------------|-----------------------|
| 5-Chloro-2-methyl-3(2H)-isothiazolone, mixture with 2-methyl-3(2H)-isothiazolone | 0.7                   |

Mobility in soil No information available.

### SECTION 13: Disposal considerations

Waste chemicals Dispose of in accordance with local regulations. Dispose of waste in accordance with environmental legislation.

Contaminated packaging Do not reuse empty containers.

### SECTION 14: Transport information

IMDG Not regulated  
Transport in bulk according to Annex II of MARPOL and the IBC Code No information available

IATA Not regulated

JT/T 617 Not regulated

**Special precautions for user**  
Please refer to the applicable dangerous goods regulations for additional information

### SECTION 15: Regulatory information

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

#### **National regulations**

**Law of the People's Republic of China on Prevention and Control of Occupational Diseases**

Catalogue of occupational hazard factors: Not applicable.  
Catalogue of occupational diseases: Not applicable.

**Regulations on the Control over Safety of Hazardous Chemicals**

Inventory of hazardous chemicals Not applicable.

GB 18218-2009 Identification of major hazard installations for dangerous chemicals Not applicable

**List of hazardous chemicals under priority management** Not applicable

**Regulations on Labour Protection in Workplaces Where Toxic Substances Are Used**

Inventory of highly toxic goods Not applicable

**Regulations for Environmental Management on the First Import of Chemicals and the Import and Export of Toxic Chemicals**

List of toxic chemicals severely restricted for import and export in China Not applicable

**Measures for the Environmental Management of New Chemical Substances**

**IECSC - China Inventory of Existing Chemical Substances** Contact supplier for inventory compliance status.

#### **International Regulations**

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

The Rotterdam Convention Not applicable

## SECTION 16: Other information

Prepared By Bio-Rad Laboratories, Environmental Health and Safety

Revision date 18-Mar-2025

Revision Note Reviewed existing information and made minor updates.

### Abbreviations and acronyms

Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | Sk*  | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

### Key literature references and sources for data used to compile the SDS

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
Environmental Protection Agency  
Acute Exposure Guideline Level(s) (AEGl(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
Japan GHS Classification  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
U.S. National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
RTECS (Registry of Toxic Effects of Chemical Substances)  
World Health Organization

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**